

- Let $\mathbb{Z}_n := \{0, 2, \dots, n-1\}$ and define

$$U_n := \left\{ \exp\left(\frac{2k\pi i}{n}\right) : k \in \mathbb{Z}_n \right\},$$

then $(U_n, \cdot)$ is the n -th root of unity group.

- Cancellation laws:** $\forall a, b, u, v \in G$, if $au = av$, then $u = v$; if $ub = vb$, then $u = v$.
- $\text{GL}_n(\mathbb{F}) := \{\mathbf{A} \in \mathcal{M}_{n \times n} : \det(\mathbf{A}) \neq 0\}$.
- $\forall$ finite Σ with $|\Sigma| = n$, define $S_{n, \Sigma}$ to be all bijections from $\mathbb{Z}_n^+$ to Σ , then $|S_{n, \Sigma}| = n!$. (Induction with disjoint union on image of $k+1$.) $\implies |S_n| = n!$.
- S_n is non-Abelian iff $n \geq 3$ ($\implies : \exists ab \neq ba$ but neither is identity; $\Leftarrow : (12)$ and (13) don't commute)
- Dihedral:** $r^n = s^2 = 1, rs = sr^{-1}$.
- S_n is generated by transpositions (induction, define τ_σ to swap $k+1$ and $\sigma(k+1)$ if they are unequal and id otherwise. $\tau_\sigma \circ \sigma \in S_k$ and $\tau_\sigma = \tau_\sigma^{-1}$.)
- S_n is generated by $(1, k)$'s (every cycle is $\prod_{i=2}^m (1, i)$, so cycle decomp then transposition decomp).
- $S_n = \langle (i, i+1) \text{'s} \rangle ((1n) = (n-1, n)(1, n-1)(n, n-1))$
- $S_n = \langle \{(12), (12 \dots n)\} \rangle$
- reversal:** $a < b$ but $\sigma(a) > \sigma(b)$. $R(\sigma)$ is all reversals.
- An m -cycle has exactly $(m-1)$ reversals.
- $\forall H \leq G$ s.t. G finite, $\exists$ maximal $M \leq G$ s.t. $H \subseteq M$ (use finite family of subgrps containing H)

Homomorphism

- Sign:** $\epsilon(\sigma) = (-1)^{|R(\sigma)|}$. even and odd permutations.
- Injective iff $\ker = 1$, surjective iff $\text{im} = H$.
- Alternating group:** $A_n := \ker \epsilon$. $A_1 = S_1, |A_n| = \frac{n!}{2}$.
- Let $\theta: \Delta \rightarrow \Omega$ be a bijection, then $\varphi_\theta: S_\Delta \rightarrow S_\Omega$ defined by $\varphi_\theta(\sigma) = \theta \circ \sigma \circ \theta^{-1}$ is an isomorphism ($\varphi_{\theta^{-1}} = \varphi_\theta^{-1}$) $\implies \varphi_{\theta \circ \eta} = \varphi_\theta \circ \varphi_\eta, \varphi_{\text{id}_\Delta} = \text{id}_{S_\Delta}$.

Subgroups

- Universal property:** if $H \leq G$, $\forall$ homomorphism $\varphi: G \rightarrow H$, $\exists!$ homomorphism $\psi: G \rightarrow K$ where $\text{im } \varphi \leq K \leq H$ such that $\varphi = i_K \circ \psi$.

- Subgroup criterion:** if $H \subseteq G$, then $H \leq G$ iff $H \neq \emptyset$ and $xy^{-1} \in H$ for all $x, y \in H$.
- Every subgroup of $\mathbb{Z}$ is $n\mathbb{Z}$ (n is min +int in subgroup).
- Finite subgroup criterion:** if finite $H \subseteq G$, then $H \leq G$ iff $H \neq \emptyset$ and $xy \in H$ for all $x, y \in H$.
- Subgroup is closed under intersection.
- Generated subgroup:** $\langle A \rangle := \bigcap_{A \subseteq H \leq G} H$
- $\langle A \rangle$ is all products of A 's elems (use subgrp criterion).

Cyclic groups

- Residue class:** $\bar{a} := a + n\mathbb{Z} = \{a + kn : k \in \mathbb{Z}\}$
- Reduction-modulo- n :** $\pi(a) = \bar{a}$ uniq s.t. $\pi(1) = \bar{1}$
- Universal property of $\mathbb{Z}$:** $\forall G$ and $\forall x \in G$, $\exists!$ homomorphism $\varphi: \mathbb{Z} \rightarrow G$ s.t. $\varphi(1) = x$. $\mathbb{Z}/n\mathbb{Z}$ is analogous.
- $\mathbb{Z}/n\mathbb{Z}: \{\bar{x} : \exists \bar{y} \text{ s.t. } \overline{xy} = \overline{yx} = \bar{1}\} = \{\bar{a} : \gcd(a, n) = 1\}$
- $(x^a)^b = x^{ab}$ (let $\psi(1) = x$ and $\varphi_k(n) = kn$, then $(\psi \circ \varphi_k)(1) = x^k$. $(x^a)^b = (\psi \circ \varphi_a)(b) = \psi(1)^{ab}$)
- $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \rangle$ is of order n .
- Fix $x \in G$ and let $\varphi: \mathbb{Z} \rightarrow G$ be the unique homomorphism with $\varphi(1) = x$, then $\text{im } \varphi = \langle x \rangle \leq G$ and

$$\ker \varphi = \begin{cases} n\mathbb{Z} & \text{if } |x| = n \in \mathbb{Z}^+ \\ \{0\} & \text{otherwise} \end{cases}.$$

($\ker \varphi \subseteq n\mathbb{Z}$ by division algo)

- $|x| = \infty \implies |\langle x \rangle| = \infty$; $|x| = n \implies |\langle x \rangle| = n$. (Let $\varphi: \mathbb{Z} \rightarrow G$ be the homomorphism with $\varphi(1) = x$. Last x^N so $x^{N+1} = x^k$ for smaller k . $\varphi(N+1-k) = 1 \implies N+1=k$; $|x| = n \implies x^{nk} = \varphi(nk) = 1 \implies x^r$'s are pair-wise distinct)
- All infinite cyclic groups $\cong \mathbb{Z}$
- Every subgroup of an infinite cyclic group is $\langle x^n \rangle$ for $n \in \mathbb{N}$ (φ unique isomorphism with $\varphi(1) = x$, $\psi(K) = \varphi[K]$ is bijective, every subgroup of $\mathbb{Z}$ is $n\mathbb{Z}$).
- $\forall$ group, if $x^a = x^b = 1$, then $x^{\gcd(a,b)} = 1$ (by Bezout).
- $\forall$ group and $x \in G$, $\exists n \in \mathbb{Z}$ s.t. $x^n = 1$ iff $|x| \mid n$.
- $\forall$ group G with $x \in G$ s.t. $x^n = 1$ for some $n \in \mathbb{Z}$. Let $\varphi: \mathbb{Z}/n\mathbb{Z} \rightarrow G$ be the unique homomorphism s.t. $\varphi(\bar{1}) = x$, then $\text{im } \varphi = \langle x \rangle \leq G$, $\ker \varphi = \langle |x| \rangle \leq \mathbb{Z}/n\mathbb{Z}$

- Every finite cyclic group $\cong \mathbb{Z}/n\mathbb{Z}$.
- Every subgrp of an order- n cyclic group is $\langle x^a \rangle$ for $a \mid n$
- Every subgroup of a cyclic group is cyclic.
- Let $G := \langle x \rangle$ be a cyclic group.
 - If G is infinite, then $G = \langle x^a \rangle$ iff $a = \pm 1$.
 - If G is of order n , then $G = \langle x^a \rangle$ iff $\gcd(a, n) = 1$.
- $\{a/2^n : a, n \in \mathbb{Z}\} \leq \mathbb{Q}$ is not cyclic.

Quotient groups

- $g_1 H = g_2 H \iff g_i \in g_j H \iff g_2^{-1} g_1, g_1^{-1} g_2 \in H$.
- $|H \backslash G| = |G/H|$ ($f: H \backslash G \rightarrow G/H$ by $f(X) = X^{-1} := \{x^{-1} \in G : x \in X\}$, $g: H/G \rightarrow H \backslash G$ by $g(X) = X^{-1}$ is inverse)
- $H \approx gH \approx Hg$
- Lagrange's Theorem:** $|G| = |G:H| |H|$.
- $\langle x \rangle \leq G \implies |x| \mid |G|$ for finite $G \implies x^{|G|} = 1$.
- Finite Abelian not divisible ($\forall a \exists x$ s.t. $1 = x^{|G|} = a$)
- A prime-order group is cyclic.
- $(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : a \leq n \text{ and } \gcd(a, n) = 1\}$ ($\supseteq$: Bezout $\implies \overline{sa} = sa + tn = \bar{1}$. $\subseteq$: $x = \bar{a}, y = \bar{b}$ s.t. $xy = 1 \implies n \mid ab - 1 \implies ab - kn = 1$)
- $\forall G, C_G(a) = N_G(\{a\})$ for all $a \in G$; $C_G(A) = \bigcap_{a \in A} C_G(a) \subseteq N_G(A)$; $Z(G) = C_G(G) = \bigcap_{a \in G} C_G(a)$; $N_G(G) = G$; $C_G(1) = G$.
- G Abelian $\iff Z(G) = G \iff C_G(a) = G \forall a \in G \iff C_G(A) = N_G(A) = G \forall A \subseteq G$
- $C_G(A), N_G(A) \leq G$.
- Normal subgroup:** $N \trianglelefteq G$ iff $gNg^{-1} \subseteq N \forall g \in G$.
- S_3 is non-Abelian but $\{1, (123), (132)\}$ is normal s.t. S_3/N is Abelian.
- G finite, $N \trianglelefteq G$ and $\gcd(|N|, |G:N|) = 1 \implies N$ is unique subgrp of ord $|N|$ (use $|HN|$ and Lagrange to prove $H = H \cap N$. Consider division to force it)
- $\mathbb{Q}$ is the only subgrp of $\mathbb{Q}$ of finite idx ($|q+H| \mid |\mathbb{Q}:H| = n \implies nq + H = H \implies n\mathbb{Q} \subseteq H$)
- $|G| = pq$ for primes $p, q \implies G$ Abelian or $Z(G) = 1$ (Lagrange: $|Z(G)| \mid pq$, then 4 cases)

- $|G:H| = m, |G:K| = n \implies |G:H \cap K| \in [\text{lcm}(m,n), mn]$ (prove $g(H \cap K) = gH \cap gK$ for mn ; use nested Lagrange for lcm)
- **Quotient group:** $(G/N, \cdot)$ where $N \trianglelefteq G$. $1_{G/N} = N$.
- **Universal property:** Let $N \trianglelefteq G$. For any group H and any homomorphism $\varphi: G \rightarrow H$ with $N \leq \ker \varphi$, $\exists! \bar{\varphi}: G/N \rightarrow H$ such that $\varphi = \bar{\varphi} \circ \pi$ ($\bar{\varphi}(\xi) = \varphi(g)$ iff $\pi(g) = \xi$).
- **1st isomorphism thm:** For any homomorphism $\varphi: G \rightarrow H$, $\exists!$ isomorphism $\tilde{\varphi}: G/\ker \varphi \rightarrow \text{im } \varphi$ s.t. $\varphi = i_{\text{im } \varphi} \circ \tilde{\varphi} \circ \pi$ is the canonical factorisation of φ (prove injection by $\ker = 1$).
- $\forall$ homomorphism $\varphi: G \rightarrow H$: $|\text{im } \varphi| = |G: \ker \varphi|$; G finite $\implies |\text{im } \varphi| \mid |G|$; G, H finite $\implies |\text{im } \varphi| \mid |H|$.
- $M, N \trianglelefteq G$ s.t. $G = MN \implies G/(M \cap N) \cong (G/M) \times (G/N)$ ($\varphi: G \rightarrow (G/M) \times (G/N)$ with quotient comp and apply 1st thm)
- **2nd isomorphism thm:** if $H, K \leq G$ and $K \trianglelefteq G$, then $H/(H \cap K) \cong HK/K$. ($K \trianglelefteq HK$. Define $\varphi = \pi_{HK} \circ i_H: H \rightarrow HK/K$. Use 1st isomorphism thm to get isomorphism $\tilde{\varphi}$. First prove it's surjective and $\text{im } \varphi = HK/K$. Then $\ker \varphi = H \cap K$ because $hk \notin K$ for $h \in K - H$.)
- **3rd isomorphism thm:** if $N, K \trianglelefteq G$ s.t. $N \leq K$, then $K/N \trianglelefteq G/N$ and $G/K \cong (G/N)/(K/N)$. ($N \trianglelefteq K, K/N \leq G/N \forall gN \in G/N$ and $k \in K, gkg^{-1} \in K$ and $(gN)(kN)(g^{-1}N) = (gkg^{-1})N \in K/N$. $\forall k \exists k'$ s.t. $k = gk'g^{-1}$ so $kN = (gN)(k'N)(g^{-1}N) \implies K/N \trianglelefteq G/N$. Let $\varphi = \psi \circ \phi$ s.t. $\psi: G \rightarrow G/N$ and $\phi: G/N \rightarrow (G/N)/(K/N)$ are quotient, then φ is surjective and $\ker \varphi = K$. Now use 1st thm.)
- **Lattice point isomorphism thm:** $N \trianglelefteq G, \pi: G \rightarrow G/N$ be quotient, $\mathcal{H}$ be family of all subgrps containing N and $\mathcal{Q}$ be family of all subgrps of G/N , $\implies \varphi: \mathcal{H} \rightarrow \mathcal{Q}$ by $\varphi(H) := \pi[H]$ is bijective s.t. $H \trianglelefteq G \implies \varphi(H) \trianglelefteq G/N$; $X \trianglelefteq G/N \implies \varphi^{-1}(X) \trianglelefteq G$.

Group actions

- $\sigma_g = a \mapsto ga$ is bijective and $g \mapsto \sigma_g$ is a homomorphism (prove $\sigma_{g^{-1}} = \sigma_g^{-1}$).
- **Action homomorphism / perm representation:** $\varphi_\alpha(g)(a) = \alpha(g, a)$. Always injective

- $\forall$ homomorphism $\varphi: G \rightarrow S_A, \alpha_\varphi = (g, a) \mapsto \varphi(g)(a)$ is a group action.
- $\psi: H \rightarrow G$ homomorphism, $\varphi: G \rightarrow S_A$ perm representation, $\alpha_\psi(h, a) = \varphi(\psi(h))(a)$ is a group action.
- **Cayley's Thm:** every group $\cong$ some subgroup of some symmetric group.
- $\alpha \mapsto \varphi_\alpha$ is bijective (prove $\varphi \mapsto \alpha_\varphi$ is inverse).
- if p is min prime factor of $|G|, |G:H| = p \implies H \trianglelefteq G$. (left $\times$ action α of G on G/H . 1st thm: $\exists K \subseteq S_{G/H}$ s.t. $G/\ker \varphi_\alpha \cong K$. Lagrange: $|G: \ker \varphi_\alpha| \mid p!$. Prove $\ker \varphi_\alpha \leq H \leq G$ and $|H: \ker \varphi_\alpha| = \frac{|G: \ker \varphi_\alpha|}{|G:H|} \Bigg| \frac{p!}{p} = (p-1)!$. Lagrange: $|H: \ker \varphi_\alpha| \mid |G| \implies |H: \ker \varphi_\alpha|$ no prime factor.)
- $\forall \sigma \in S_A$, define $\mathcal{P}(\sigma): \mathcal{P}(A) \rightarrow \mathcal{P}(A)$ by $\mathcal{P}(\sigma)(U) = \sigma[U]$, then $f(\sigma) = \mathcal{P}(\sigma)$ is a homomorphism.
- $\alpha((\sigma_A, \sigma_B), f) = \sigma_B \circ f \circ \sigma_A^{-1}$ is a group action of $S_A \times S_B$ on $\text{Maps}(A, B)$
- **Stable:** $A_0 \subseteq A$ is stable under α if $\alpha[G \times A_0] \subseteq A_0$.
- A_0 is stable under $\alpha \iff \varphi_\alpha[G] = \text{Perm}(A)_{A_0} := \{\sigma \in S_A: \sigma[A_0] = A_0\} \iff \sigma_g(a) \in A_0 \text{ for any } g \in G \text{ and } a \in A_0 \iff \varphi_\alpha|_{G \times A_0}$ is an action homomorphism
- **Orbit:** $G \cdot a$ is all b s.t. $\exists g$ s.t. $b = ga$.
- **Transitive action:** only one orbit
- **Stabiliser:** $G_a := \{g \in G: ga = a\} \subseteq G$
- $f(gG_a) = ga$ is a bijection $\implies |G: G_a| = |G \cdot a|$
- **Fixed point:** $ga = a \forall g \in G$. A^G is all fixed points.
- $|A| = |A^G| + \sum |G: G_{a_i}|$ where $G \cdot a_i$'s non-trivial orbits
- Conjugate action: fixed points $Z(G)$, orbit $C_G(g_i)$
- **p -group:** $|G| = p^\alpha$ for some $\alpha \geq 1$.
- **Sylow p -subgroup:** maximal p -subgroup
- Let $|G| = p^\alpha m, \alpha \geq 1$ and $p \nmid m$, then $H \leq G$ is a Sylow p -subgroup iff $|H| = p^\alpha$ or $\gcd(|G:H|, p) = 1$.
- **Basic lem:** H p -group act on $A \implies |A^H| \equiv |A| \pmod p$.

- If H is a p -subgroup of a finite group G , then $|N_G(H):H| \equiv |G:H| \pmod p$ (use left $\times$ action of H on G/H and prove $(G/H)^H = N_G(H)/H$).
- Centre of any p -group is non-trivial
- **Cauchy's Thm:** $\forall$ finite $G, \forall$ prime $p \mid |G|, \exists g \in G$ s.t. $|g| = p$ ($\sigma_0 = (1 \cdots p), \langle \sigma_0 \rangle \cong \mathbb{Z}/p\mathbb{Z}$. α acts on G^p with σ_0 . Show $A := \{\prod g_i = 1\}$ stable. Orbit-decomp $\implies |A| = |A^{\langle \sigma_0 \rangle}|$. Compute $|A|$ to show A non-trivial)
- **Sylow's Theorems:**
 1. $\forall$ finite G and prime factor p with multiplicity α of $|G|, \exists$ Sylow p -subgrp of ord p^α (show $\exists p$ -subgrp $\forall k$. Lagrange: $p \mid |G:H|$. $H \trianglelefteq N(H)$. Basic lem: $p \mid |N(H):H|$. Cauchy: $\exists |\langle gH \rangle| = p$ and each coset has p^k elems. $\pi^{-1}(K)$ has ord p^{k+1})
 2. $\forall P, Q \in \text{Syl}_p(G), \exists g \in G$ such that $Q = gPg^{-1}$ (w.r.t left $\times$ action, show $(G/P)^Q = \{gP: g^{-1}Qg \subseteq P\}$. Q maximal so $\subseteq$ is $=$. Show it's not $\emptyset$)
 3. $\forall G$ s.t. $|G| = p^\alpha m$ with $\alpha \geq 1$, and $p \nmid m \geq 1$: $n_p(G) \mid m; n_p \equiv 1 \pmod p; n_p = |G: N_G(P)|$ for any Sylow p -subgroup (by 2: Syl_p transitive under conj so $n_p = |\text{Syl}_p| = |G \cdot P| = |G/G_P| = |G: G_P|$. $\forall P$ Sylow, $G_P = N(P)$. Show Syl_p^P under conj is $\{Q \in \text{Syl}_p: P \subseteq N(Q)\}$. Maximality of P, Q : both Sylow in $N(Q)$. by 2: $\exists \nu \in N(Q)$ s.t. $P = \nu^{-1}Q\nu = Q$ so $\text{Syl}_p^P = \{P\}$. Basic lem: $n_p \equiv 1 \pmod p$. Lagrange: $n_p \mid m$)
- Every p -subgroup is in some Sylow p -subgroup
- **Simple group:** non-trivial and 1 and G are only normal subgroups.
- Every group of order < 60 is either Abelian simple and $\cong \mathbb{Z}/p\mathbb{Z}$ of prime order p , or not simple.
- p -group with order $> p$ is not simple (centre is normal)
- Any finite group of order $p^\alpha q^\beta$ where p and q are primes, $\alpha, \beta \geq 1$, and $p^k \not\equiv 1 \pmod q$ for any $k = 1, 2, \dots, \alpha$ is not simple (Sylow 3rd: $n_q \equiv 1 \pmod q$ and $n_q \mid p^\alpha$ so only 1 Sylow subgrp. Sylow 2nd: $gSg^{-1} = S$ so G not simple)